

CHAINLAB

Blockchain Fundamentals

A beginner-friendly learning guide

FREE BLOCKCHAIN EDUCATION RESOURCE

This guide is designed for learning and revision. It explains foundational concepts and highlights practical limitations and safety considerations.

Learning objectives

- Explain what a blockchain is and why it is used.
- Describe blocks, transactions, nodes, consensus and cryptographic hashes.
- Compare common blockchain designs and identify limitations.

1. What is a blockchain?

- A blockchain is a shared record of transactions maintained by a network of computers. Records are grouped into blocks, and each block refers cryptographically to the previous one. This makes later alteration detectable.
- A blockchain is not automatically private, anonymous, fast, or free. Its properties depend on the protocol, governance, network design and application.

2. Core components

- Transactions: proposed state changes, such as transferring a digital asset.
- Blocks: batches of transactions plus metadata, including a reference to the previous block.
- Nodes: computers that store, validate, relay or produce blocks, depending on their role.
- Cryptographic hashes: fixed-length outputs used to identify data and detect changes.
- Consensus: rules by which participating nodes agree on an accepted history.

3. How a transaction is processed

- A user creates and signs a transaction with a private key.
- The transaction is broadcast to the network and checked against protocol rules.
- A block producer or validator includes eligible transactions in a block.
- Nodes independently verify the block and update their view of the ledger.
- Finality describes when a transaction is considered settled; its strength and timing vary by network.

4. Consensus overview

- Proof of Work uses computational work to influence block production and chain selection. It can require substantial energy and hardware.
- Proof of Stake uses stake and protocol rules to select or authorize validators. Designs differ in penalties, decentralization and finality.
- Neither label alone guarantees security. Network participation, incentives, client diversity and implementation quality matter.

5. Strengths and trade-offs

- Potential strengths include shared verification, auditability, programmable rules and reduced dependence on one record keeper.

- Trade-offs include throughput constraints, fees, key-management risk, governance disputes, software bugs, privacy concerns and difficult recovery after mistakes.

Review questions

- What information does a block reference from the previous block?
- What is the difference between a node and a validator?
- Name two benefits and three limitations of blockchain systems.

Educational notice: This material is for general education only. It is not financial, legal, tax or investment advice. Verify current network and project details before acting.